

Week 8 Weekly Report

Week 8 integrated the complete project into a self-contained chooser pricing application. The tool exposes the Week 4 BSM model, the selected Week 6 volatility-plus-BSM route, the direct-price benchmark, unit-labelled Greeks, sensitivity analysis, performance evidence, and live-data status.

1 Work Completed

- Bundled the original validated BSM engine and trained models, with a complete 2024-12-30 offline feature context and independent runtime source caches.
- Implemented dual pricing with BSM and the selected Random Forest volatility-plus-BSM route.
- Added a 90% proxy-error band calibrated from purge-aware out-of-fold training predictions and evaluated on the test period.
- Added input-domain warnings when live spot, rate, volatility, or VIX falls outside the Week 5 model range.
- Built a four-tab Streamlit dashboard for pricing, sensitivity, performance, and limitations.
- Ran notebook integration checks, online source-refresh tests, feature-parity and fallback tests, and a Streamlit page test. Added automatic refresh without retraining the models.

2 Deliverables

- Week8_Final_Integration_and_Validation.ipynb
- self-contained pricing_tool folder
- dual-pricing and stress-test outputs
- deployment readiness and submission checklist
- 5–10 minute demo script
- Final Project Report
- Final Presentation

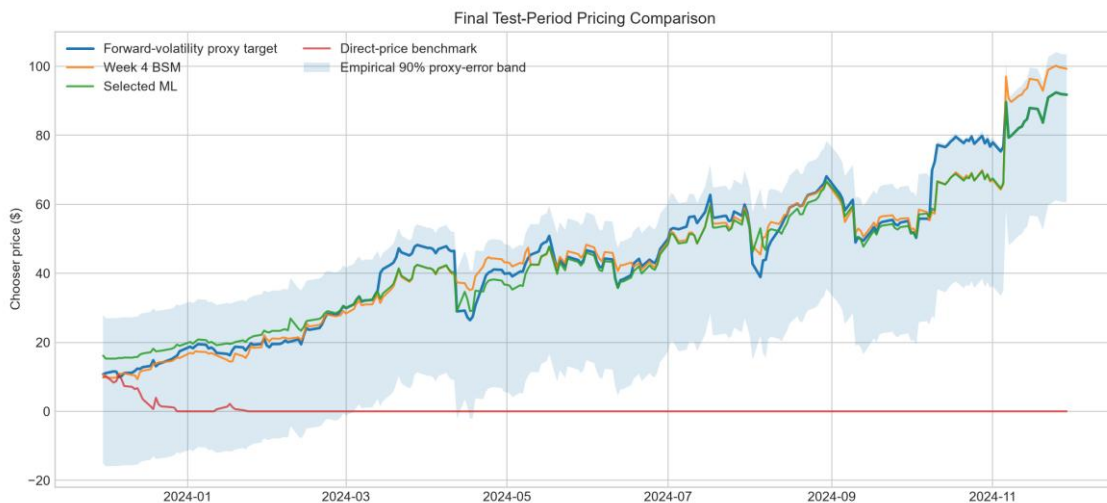
3 Progress Summary

All eight final integration checks passed. The unchanged nominal 90% proxy-error band covered 99.60% of the 252 test observations with mean width \$42.8744. The offline example now aligns every feature to 2024-12-30. Online refresh rebuilds the original 17 volatility features and 19 direct-price features at a common completed date, automatically every 15 minutes while the session is active.

4 Experimental Results

Route or interval	MAE	RMSE or coverage	R ² or width
Week 4 BSM	3.5698	4.8706	0.9460
Approach 1 RF vol plus BSM	3.1384	4.3203	0.9575
Approach 2 direct price	45.0275	49.9891	-4.6915
Empirical 90% band	—	99.60% test coverage	42.8744 mean width

Figure 1 Final held-out pricing comparison



5 Issues Encountered

- The bundled offline example uses a JPM spot of \$231.08, which exceeds the actual fitted-sample maximum of \$152.49. The application displays an ML extrapolation warning; BSM remains available without this data-driven range restriction.
- The deployable tool and README are available in the GitHub repository. Project validation, weekly preprocessing and complete market-data refresh passed on GitHub Actions on 2026-09-11. The project brief does not require public cloud hosting.
- The required narrated demo video requires the student's recording and submission. A timed script and checklist are included.

6 Plan for Next Stage

- Verify reviewer access to the GitHub repository and rehearse the locally running application.
- Record the 5–10 minute demonstration using the supplied script.
- Submit the GitHub repository with README, final report PDF, presentation deck, and narrated demo video.